MAPLE & MAIN · CUSTOMER ACQUISITION

More new customers from the same outreach budget

A propensity scorecard that finds 59% more customers at a fixed budget, and the one number to measure before spending more.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

Rank prospects by propensity now. Test what outreach causes before raising the budget.

1,247

more new customers a year when outreach reaches 40% of prospects chosen by the scorecard instead of at random. Worth $514k in first-year margin.

8x

The top tenth of prospects converts at 24%, the bottom tenth at 3%. A simple scorecard captures all of the signal the data holds.

51%

of bottom-tenth conversions must be caused by outreach for those contacts to pay back. Nobody has measured that share.

Recommendation: use the scorecard to order the outreach list today, buy lead lists filtered on profile instead of dropping sources, and run a 10% holdout for two quarters to learn how many conversions outreach really causes.

APPROACH

From a prospect list to a budget decision

Maple & Main collects about 52,000 prospects a year from five lead sources. About one in ten becomes a loyalty member within 45 days.

1. Who converts?

Three ranking methods compared on a later period: a scorecard, gradient boosting, and a look-alike index.

SQL · Python · SAS

2. What is a customer worth?

First-year margin measured for 7,456 prospects who became customers, including those who never ordered.

Linked to 2.3M orders

3. What does a ranked list buy?

Customers found at every budget level, against a random list.

Out-of-time test, 49,937 prospects

4. What is still unknown?

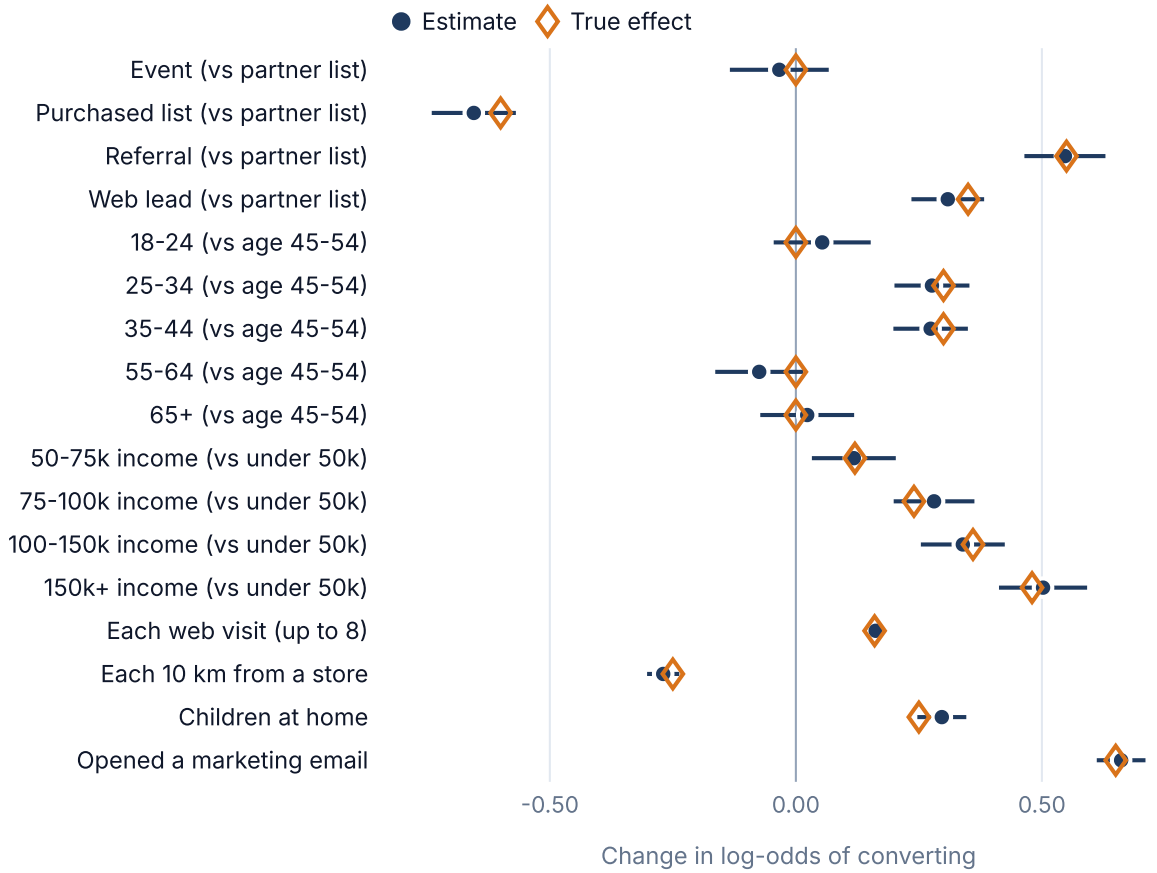
Whether outreach causes the conversion. The break-even is worked out for each score band, with the test to measure it.

Sensitivity and test design

DATA USED	CHECKS BEFORE MODELLING	HOW RESULTS WERE VALIDATED
150,000 prospects with lead source, profile and web behaviour, linked to the customer and order files	10,630 prospects removed because their 45-day conversion window was not fully observed; eight integrity checks passed	Trained on 2024 to September 2025, tested on the following year, and compared with the conversion rule built into the data

WHAT DRIVES CONVERSION

Email engagement, lead source and distance to a store matter most



Effect on the log-odds of converting, with 95% intervals. Diamonds show the true effect built into the synthetic data.

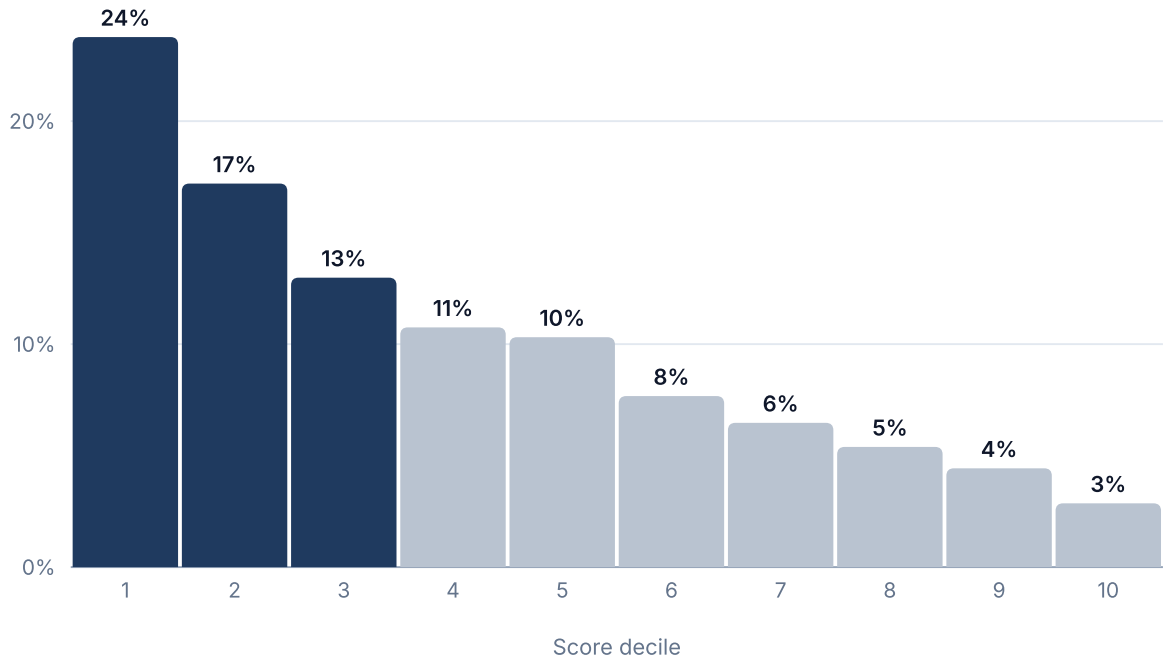
- A prospect who has opened a marketing email has about **1.9 times** the odds of converting.
- Referrals and web leads convert best. Purchased lists convert worst.
- Every 10 km from a store cuts the odds by about **24%**.
- Ages 25 to 44, households with children and higher incomes all convert more.

> All 17 estimated effects fall within their margin of error of the true value. The scorecard has learned the real mechanism, not noise.

MODEL PERFORMANCE

An AUC of 0.68 looks modest. It is also the most this data can give.

Conversion rate by score decile, prospects from October 2025 to September 2026



Decile 1 is the highest score. Average conversion is 10.2%.

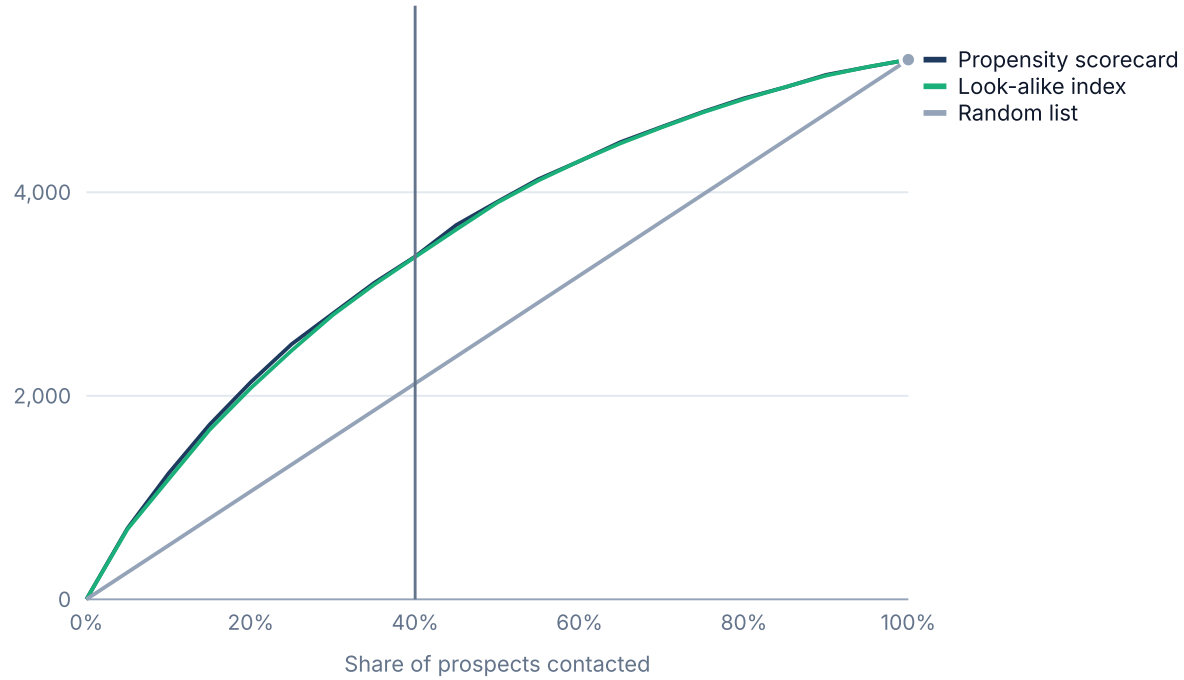
RANKING METHOD	AUC	SHARE OF ACHIEVABLE SIGNAL
Web visits only	0.58	46%
Look-alike index	0.68	97%
Gradient boosting	0.68	97%
Propensity scorecard	**0.68**	**100%**
Ceiling: the true conversion rule	0.68	100%

- Conversion is partly chance. Even a perfect model would score 0.68 here.
- The more complex model does slightly worse than the scorecard. More modelling will not help; new data might.
- Conversion has been rising, so scores read low: 9.3% predicted against 10.2% actual. The ranking is unaffected; the baseline needs a quarterly refresh.

BUSINESS CASE

At the same budget, the ranked list finds 1,247 more customers a year

New customers found, by share of prospects contacted



Prospects from the test year. The look-alike index sits almost exactly on the scorecard line.

PROSPECTS CONTACTED	RANDOM LIST	SCORECARD	EXTRA FIRST-YEAR MARGIN
20% (10,415)	1,061	2,134	$442k
40% (20,831)	**2,121**	**3,368**	**$514k**
60% (31,246)	3,182	4,306	$463k
80% (41,662)	4,242	4,923	$281k

Customers a year. Margin uses the measured $412 first-year margin per new customer.

> At 40% reach, outreach cost per new customer falls from $59 to $37. The look-alike index does as well as the scorecard, so a ranked list is possible even without records of who did not convert.

LEAD SOURCES

Profile matters more than source: a good purchased-list name beats a weak referral

Conversion rate by lead source and score band

LEAD SOURCE	BOTTOM THIRD	MIDDLE THIRD	TOP THIRD
Event	5.2%	8.2%	14.8%
Partner list	4.4%	8.8%	14.8%
Purchased list	4.1%	7.3%	14.2%
Referral	6.1%	10.3%	17.9%
Web lead	5.1%	9.5%	18.2%

Score band is the scorecard third across all prospects.

Cost to acquire a customer, by source

LEAD SOURCE	CONVERTS	COST PER LEAD	COST PER CUSTOMER
Web lead	14.7%	$8	$95
Partner list	8.8%	$5	$125
Purchased list	4.9%	$2	$162
Referral	14.9%	$20	$174
Event	8.6%	$15	$243

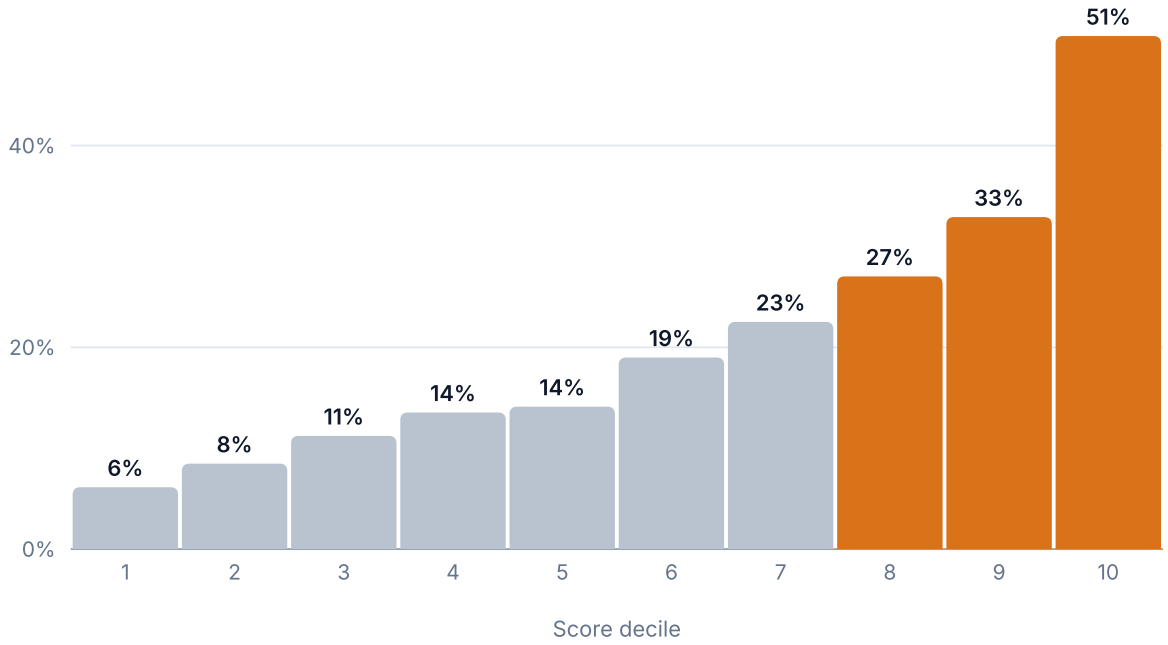
Cost per lead is an assumption, not data. Cost per customer adds $6.00 of outreach per prospect. Change both in the simulator.

> Cutting purchased lists would lose good prospects. Buying them filtered on income, age, children and distance keeps the low price and lifts conversion.

THE NUMBER NOBODY HAS MEASURED

Every contact pays back only if outreach causes the conversion

Share of conversions outreach must cause for a contact to break even



By score decile. Break-even is the $6.00 contact cost divided by conversion rate times $412 first-year margin.

Some prospects would have joined anyway. The data records who converted, not who converted *because* they were contacted. There has never been a holdout.

IF OUTREACH CAUSES	NET A YEAR, CONTACT ALL	DECILES LOSING MONEY
100% of conversions	$1872k	0
75% of conversions	$1326k	0
50% of conversions	$780k	1
25% of conversions	$234k	3

> The top deciles are safe under almost any assumption. The case for contacting the bottom third rests on a number that has not been measured.

TEST DESIGN

A 10% holdout for two quarters settles it

Smallest effect the test can detect, in conversion points

HOLDOUT SHARE	ONE QUARTER	TWO QUARTERS	ONE YEAR
5%	3.4 (33%)	2.4 (24%)	1.7 (17%)
10%	**2.5 (24%)**	**1.8 (17%)**	**1.2 (12%)**
20%	1.9 (18%)	1.3 (13%)	0.9 (9%)

80% power, 5% two-sided significance, at 52,000 prospects a year and a 10.2% conversion rate. In brackets: the smallest share of conversions caused by outreach that the test would detect.

- **Design:** hold back a random 10% of prospects from outreach for 26 weeks. Compare 45-day conversion.
- **What it answers:** whether outreach causes at least 17% of conversions, and how that varies by score band.
- **Cost:** about 2,600 prospects go uncontacted. If outreach is fully effective, that forgoes roughly $109k of first-year margin, less the contact cost saved.
- **Decision rule:** expand outreach to every score band whose measured incremental value exceeds the contact cost.

RECOMMENDATIONS

Four actions

1 **Order the outreach list by the scorecard**

It is a 17-term formula that runs in SQL. Refresh the baseline every quarter because conversion is drifting up.

$514k a year

at 40% reach

2 **Buy lead lists on profile, not just price**

Filter purchased and partner lists on income, age, children and distance to a store before buying.

Depends on list pricing

3 **Run the 10% holdout for two quarters**

Until then, treat any plan to extend outreach to low-scoring prospects as unproven.

Decides the budget

4 **Stop tuning the model; add new signals**

The scorecard already captures all the signal in these inputs. Product interest, basket intent or postal-code demographics are the way to a better ranking.

Raises the ceiling

APPENDIX

Method, assumptions and limits

Population. Prospects first contacted from 1 January 2024 to 15 September 2026. Later prospects are excluded because conversion can take 45 days.

Target. Became a loyalty member within 45 days of first contact.

Validation. Trained on 89,433 prospects to September 2025, tested on 49,937 from the following year.

Scorecard. Unregularised logistic regression on lead source, age band, income band, web visits (capped at 8), distance, children and email engagement.

Look-alike index. For each trait, the share of converters with it divided by the share of all prospects with it; a prospect's score is the product of its indices.

Customer value. Mean first-year margin of converted prospects with a full year of orders. 10% never ordered and are included at zero.

Assumed, not measured. Outreach cost of $6.00 per prospect, and cost per lead by source. Both can be changed in the simulator.

Not known. How many conversions outreach causes. All "extra customer" figures compare ranked and random lists of contacted prospects and assume outreach is what converts them.

Limits. First-year margin understates lifetime value. Higher-scoring prospects become slightly more valuable customers (about 9% between bottom and top third); a single average value is used.

Synthetic data. Maple & Main is fictional. The data was generated with a known conversion rule, which is why the model can be compared with the truth. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a model card and QC log, are in the project repository.